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Devixo Solutions

AI/ML Internship Program

Task 03

Project

Internship Domain : AI/ML

Submission Date: 25/0ct/2026

Dataset:

Source: Tesla Stock Data, from Kaggle which contains than 3,000 rows of past market records.

Target Variable: Closing stock price, also known as Close which's the main value I aim to predict.

Features: price, High price, Low price, Trading volume 5- day Moving Average (MA_5) and Lag Close (Lag_Close).

Workflow:

Data Preparation: I took care of missing values I removed duplicate records I created lag and rolling features I split the data into eighty percent for training and twenty percent for testing and I applied standard scaling.

Model Development: I trained four regression models: Random Forest, Gradient Boosting, Support Vector Regressor (SVR) and Decision Tree.

Optimization: I performed cross- validation and I used GridSearchCV to adjust the hyper- parameters of the Random Forest model.

Artifact Saving: I serialized the optimized model and the data scaler, with joblib.

Model Selection & Results:

The best model was the Random Forest Regressor. It gave the stable and accurate predictions during cross-validation. It achieved a R-squared score and a low Mean Squared Error. The Random Forest Regressor performed well in both accuracy and consistency.

Limitations:

Stock market prices change a lot. Are affected by big things outside the market. These include the state of the economy how people feel about news and events between countries. Technical tools, by themselves can’t show all of this.

Future Improvements:

Add sentiment analysis from news headlines or Twitter/X data using NLP. It will help us understand how people feel about the market.

Upgrade the architecture from machine learning to deep learning models, like LSTMs or Transformers for time- series forecasting. That will give us predictions.

Interface:

